John Cole

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Professional Summary

Software Engineer with a strong foundation in applied AI/ML research and production systems development. Experienced building embedded AI engines, ML pipelines, and multi-agent reinforcement learning systems. Currently completing an M.S. in Computer Science at the University of Maryland, with published research at AAMAS 2025 and ongoing thesis work on autonomous vehicle VLA models.

Education

University of Maryland at College Park — College Park, MD
M.S. Computer Science. Thesis Advisor: Tom Goldstein. GPA: 3.96/4.00. — *May 2026*

University of Maryland at College Park — College Park, MD
B.S. Computer Science, Mathematics. — *May 2022*

NSF IAIFI (MIT) — Cambridge, MA
Selective Program on AI and Fundamental Physics. — *August 2026*

Experience

Software Engineer — December 2022 – January 2025
Origin AI — *Rockville, MD*

- Led embedded engineering for a partnership with Deutsche Telekom serving approximately 14,000,000 customers.
- Reduced Wireless AI engine startup latency by 75% (from 20s to 5s) by refactoring core modules in C++.
- Architected an offline queuing mechanism for an on-edge AI intrusion detection system deployed across the full 14,000,000-customer base, enabling anomaly detection in home security routers without an active internet connection.

Full Stack Developer Intern — March 2020 – October 2022
Evans and Chambers Technology — *Remote*

- Built a rule-based threat detection module in Groovy/Grails with SQL querying to flag anomalies in security clearance applications, increasing identification of suspicious applications by 30% across approximately 500 applications per month.
- Applied Scrum Agile methodology to organize developer assignments and production schedules.

Software Developer Intern — September 2017 – May 2018
Johns Hopkins University Applied Physics Laboratory — *Laurel, MD*

- Engineered a maritime navigation tool in Java and Python projecting a 15% fuel reduction via optimal oceanic routing.
- Processed NOAA data and implemented pathfinding algorithms using Java, Python, ArcGIS, and QGIS.

Research Publications

- Multimodal Agentic Model Predictive Control -- Saptarashmi Bandyopadhyay, John Cole, Thomas Goldstein, David Jacobs. 24th ACM Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025.

Projects

Master's Thesis: Post-Training Autonomous Vehicle VLA Models Using Multi-Agent Rewards

- Developed mechanisms to incentivize safe driving behaviors and zero-shot coordination with other vehicles at the intersection of multi-agent AI, reinforcement learning, Vision-Language-Action (VLA) models, and autonomous driving.
- Designed reinforcement learning environments to simulate complex multi-agent vehicular interactions.

Multi-Agent Recommender Systems for Explainable Artificial Intelligence

- Applied Multi-Agent Reinforcement Learning methods to develop Recommender Systems for Explainable AI (XAI).
- Incorporated human feedback to guide and validate recommendation performance across multiple XAI scenarios.

MASIMU: Multi-Agent Interpretable Machine Unlearning

- Trained a model to selectively forget the influence of a designated forget set while preserving performance on a retain set.
- Applied Multi-Agent Reinforcement Learning and interpretability methods including LIME to remove targeted image data.

Explainable AI on Automatic Generation of Poetry — September 2023 – October 2023

- Investigated outputs of Google's Fine-tuned LAnguage Net (FLAN) when prompted with a poem, its author, and its genre.
- Examined structural patterns in generated text using XAI methods including InSeq.

Skills

Languages: Python, C++, C, Rust, TypeScript, SQL, Julia, R, MATLAB

ML / RL Frameworks: PyTorch, TensorFlow, JAX, Hugging Face, OpenCV, Gymnasium/RLlib/PettingZoo, Apache MXNet

Systems / Infra: Git, Docker, Linux, Kafka, HiveMQ, MongoDB, Jupyter, Jira